

Cosmic Dawn, Ringberg Castle, 18.06.13

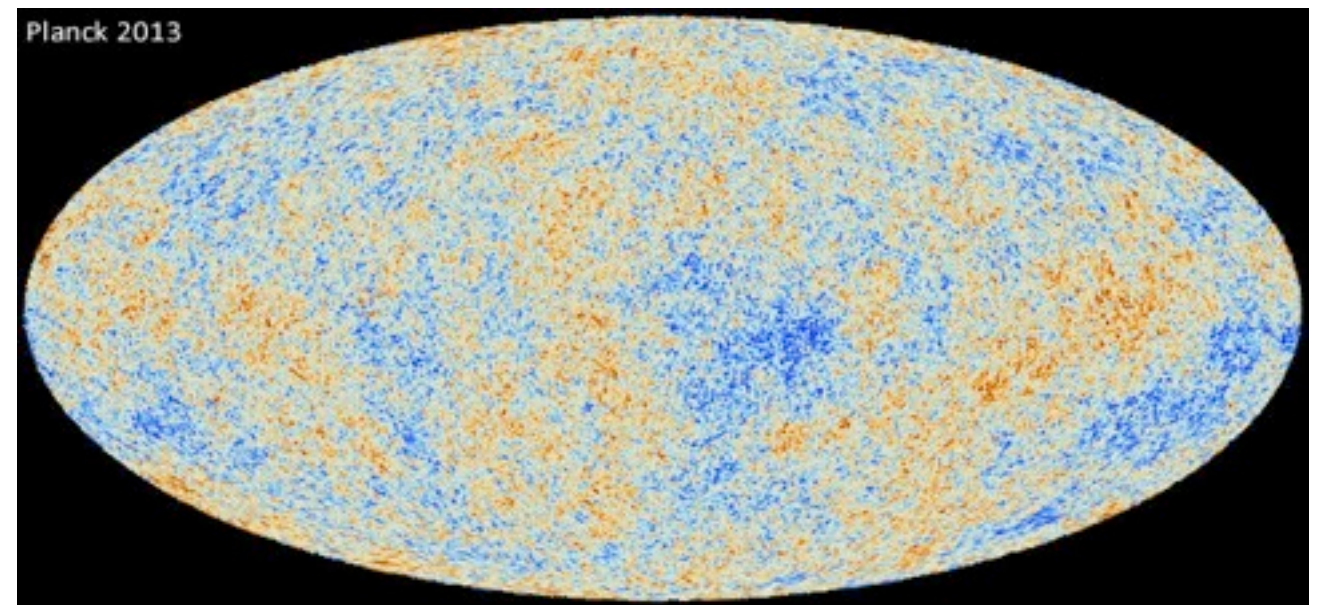
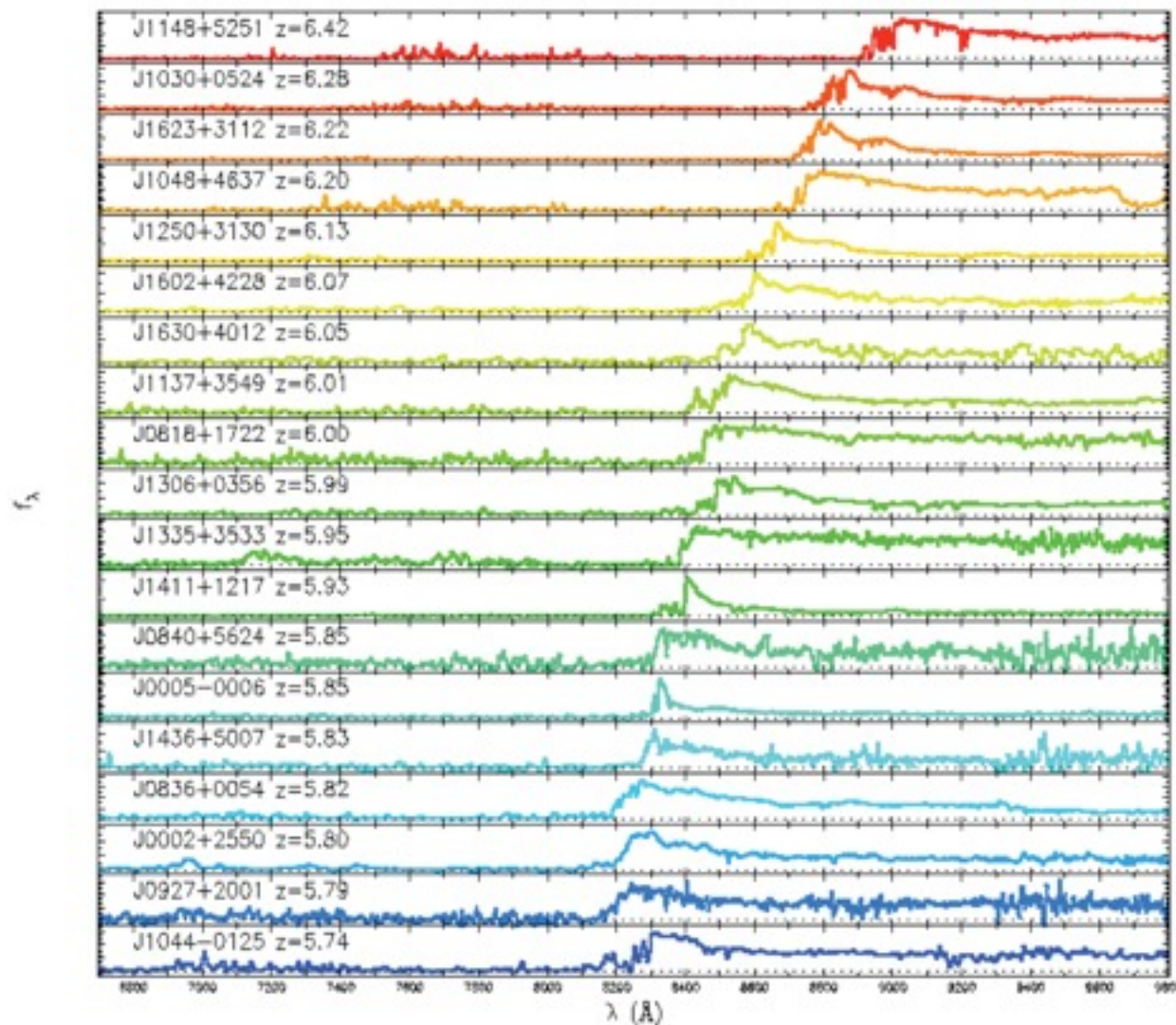
# The impact of the IGM on the visibility on Ly- $\alpha$ emitters at $z \sim 7$



Jamie Bolton

With thanks to:  
Benedetta Ciardi (MPA),  
George Becker, Martin Haehnelt (Cambridge),  
Sudhir Raskutti (Princeton), Stuart Wyithe (Melbourne)  
Steve Warren, Daniel Mortlock (Imperial)

# 1) The GP trough and CMB



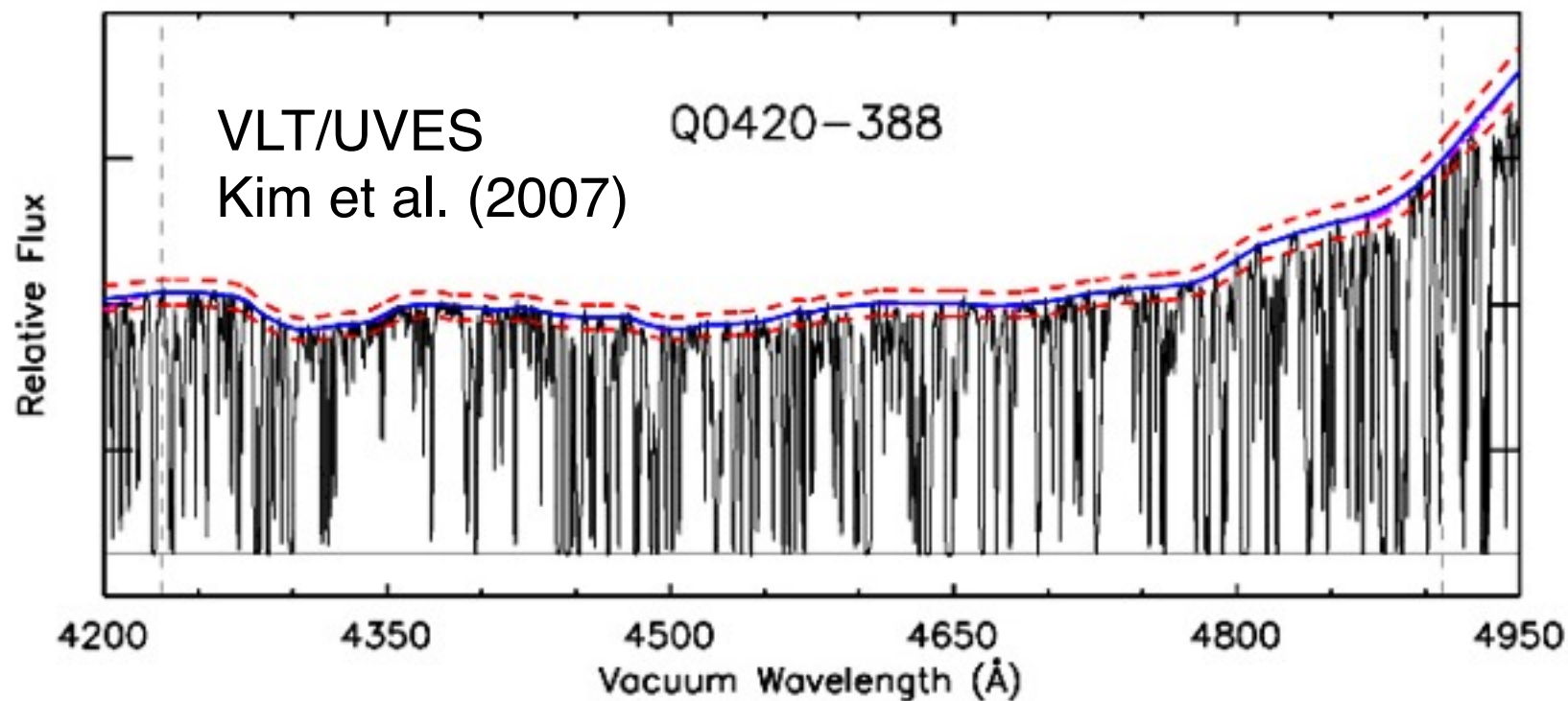
Planck Collaboration XVI (2013)

Fan et al. (2006), see also Mesinger (2010)

$$z_{\text{end}} \approx 5.5$$

$$z_{\text{reion}} = 11.3 \pm 1.1$$

## 2) How many photons at $z < 6$ ?

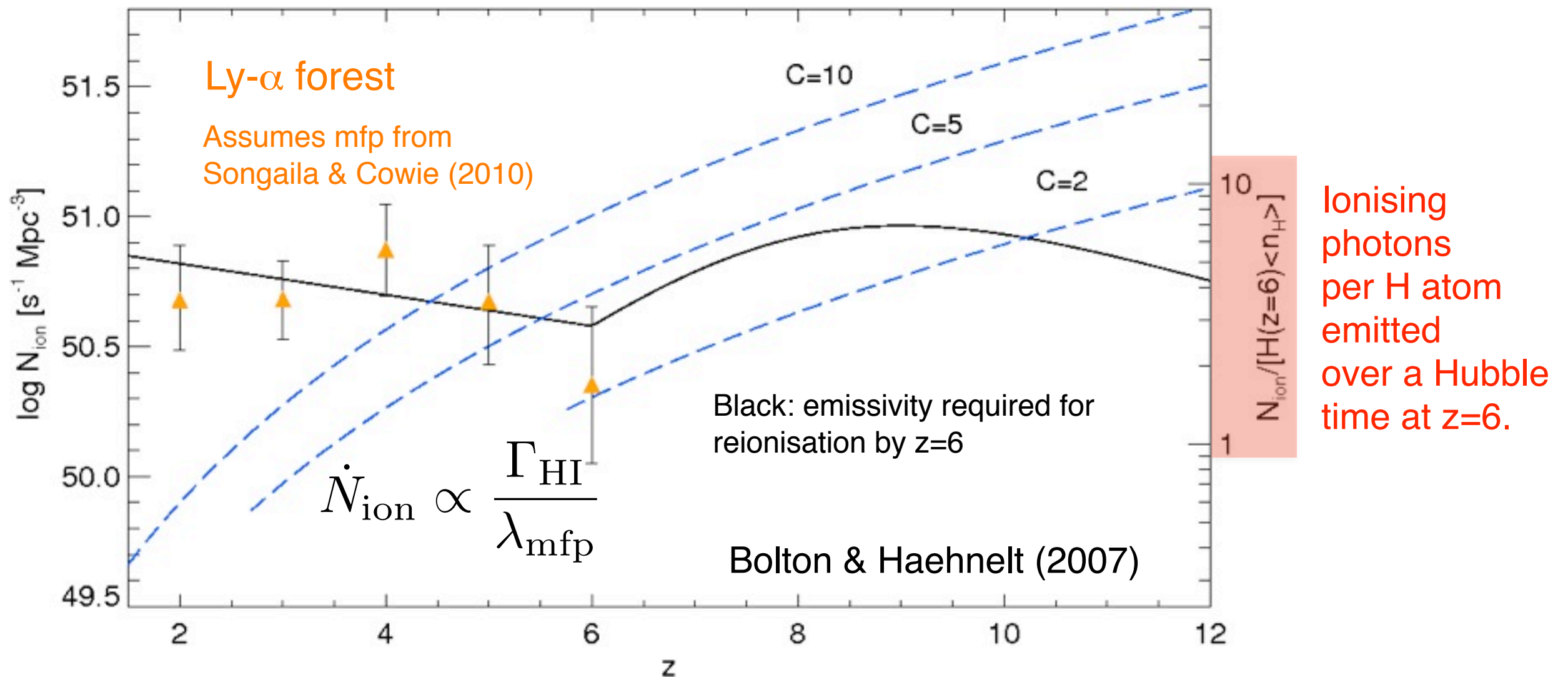


The Ly- $\alpha$  forest encodes information on the ionisation state of the IGM at  $z < 6$ .

Ly- $\alpha$  opacity  $\tau = \tau_0 \frac{(1+z)^6 (\Omega_b h^2)^2}{T_0^{0.7} H(z) \Gamma(z)} (1+\delta)^{2-0.7(\gamma-1)}$

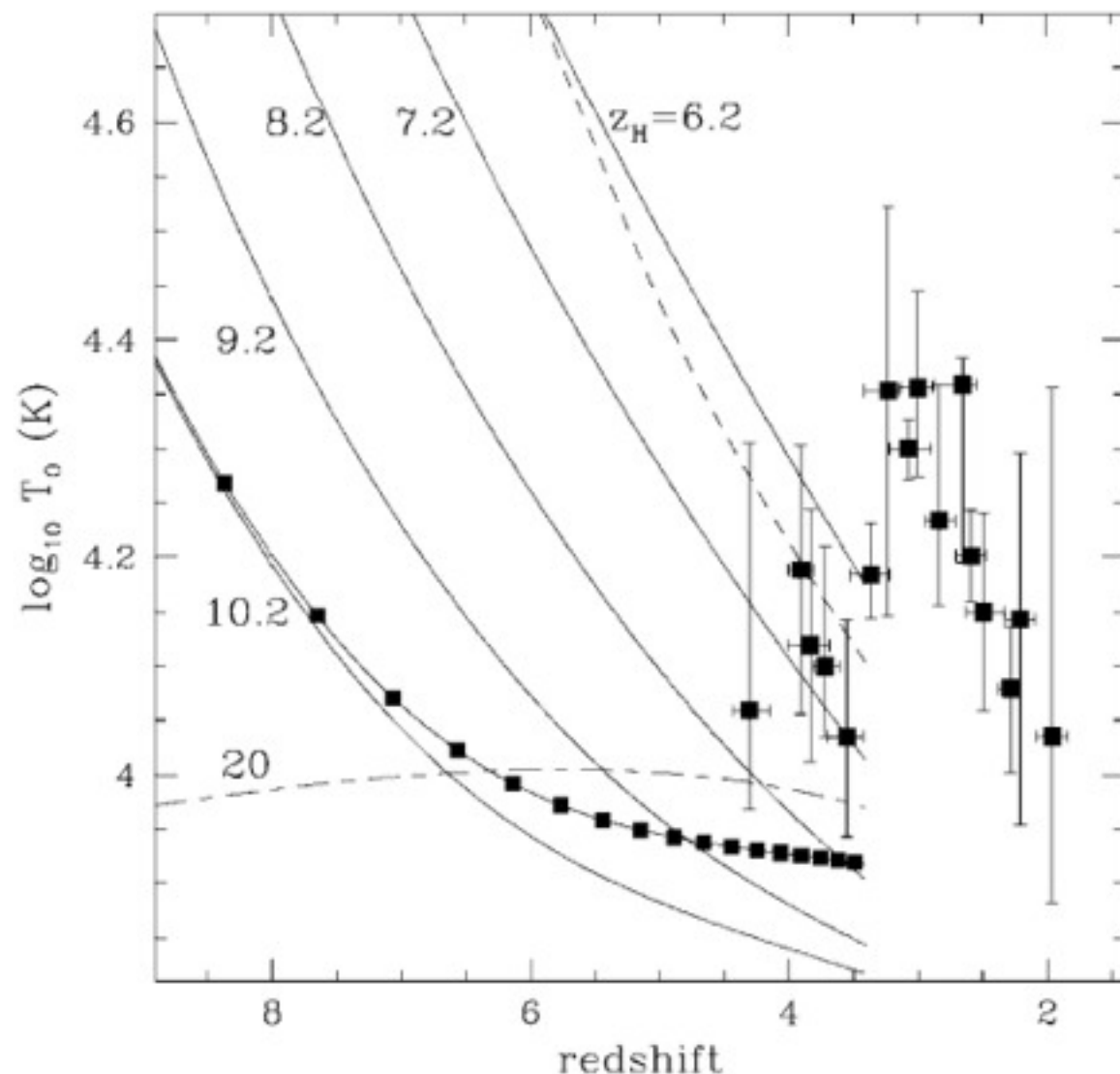
Photo-ionisation rate

## 2) How many photons at $z < 6$ ?



- Ly- $\alpha$  forest consistent with only 1-3 ionising photons per HI emitted over a Hubble time at  $z=6$  (see also Miralda-Escude 2003, Meiksin 2005, Kuhlen & Faucher-Giguere 2012);
- Ionising emissivity **must remain constant or rise at  $z > 6$**  for reionisation to complete by  $z=6$ .

### 3) The IGM temperature at $z=6$



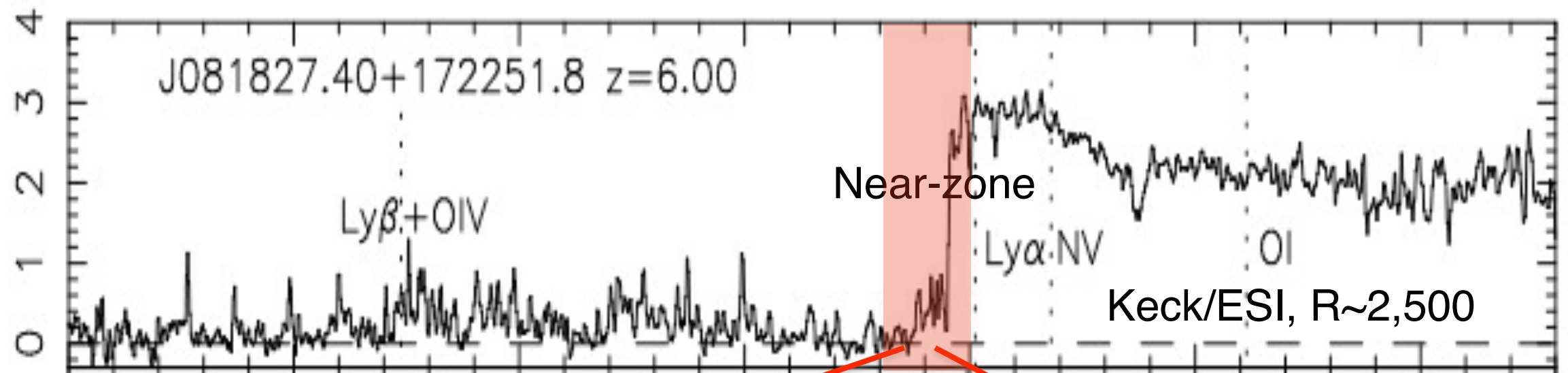
Theuns et al. (2002),  
see also Hui & Haiman (2003)

- Temperature of the low density ( $\delta < 10$ ) IGM provides indirect probe of reionisation history;
- Long cooling timescale allows IGM to retain thermal memory of reionisation;

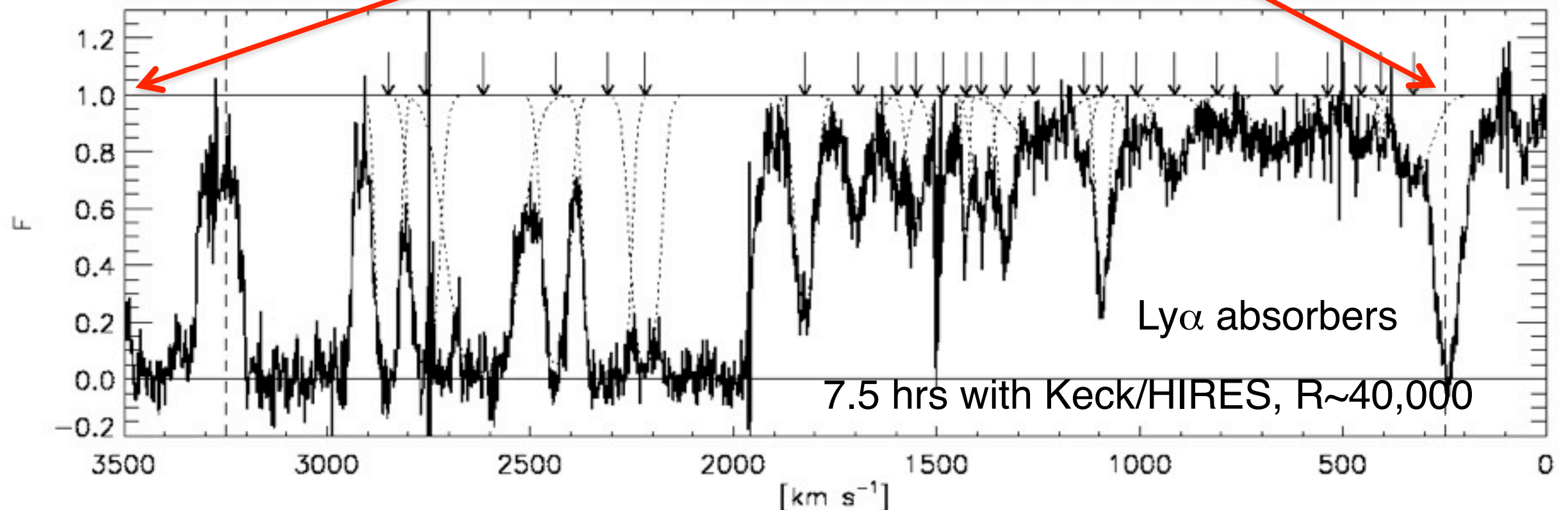
The temperature of the IGM depends on:

- 1) When the IGM was reionised (how much time available to cool?)
- 2) Spectra of the ionising sources (harder spectra = more heating).

### 3) The IGM temperature at $z=6$

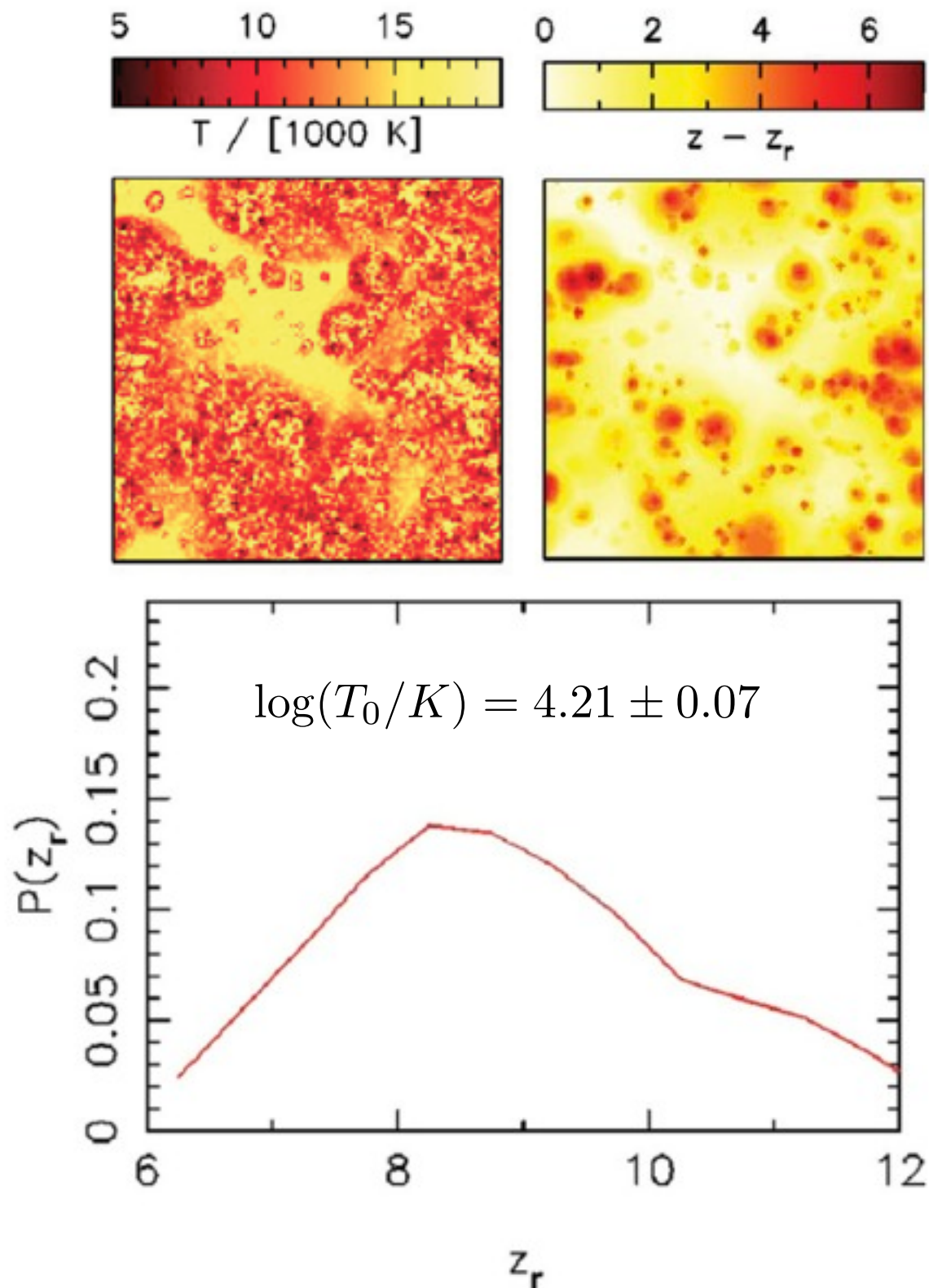


Fan et al. (2006)



Becker et al. (2007), Bolton et al. (2010, 2012)

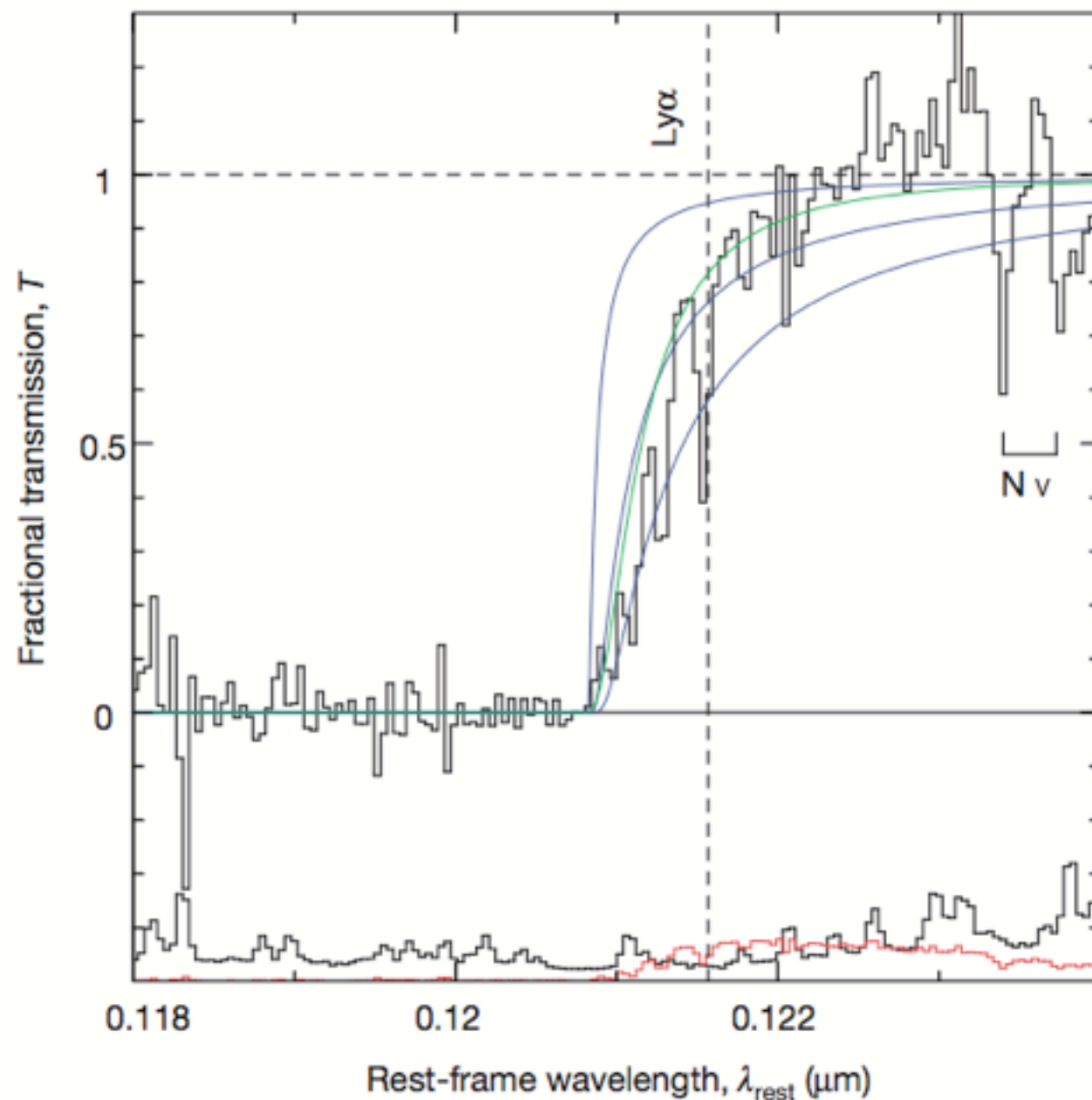
### 3) The IGM temperature at $z=6$



- Grid of large scale ( $L=100 \text{ Mpc/h}$ ), efficient “semi-numerical” models of reionisation by galaxies;
- Temperature data **inconsistent with very late end to reionisation ( $z_r < 6.5$ )** for  $f_v \sim v^{-3}$ ; observed IGM too cool at  $z=6$ ;
- **Lower limit on  $z_r$ :** higher temperatures from e.g. harder stellar sources/X-rays/significant spectral filtering favour larger  $z_r$  (more time for gas to cool).

Raskutti, JB, Wyithe et al. (2012),  
see also Theuns et al. (2002)

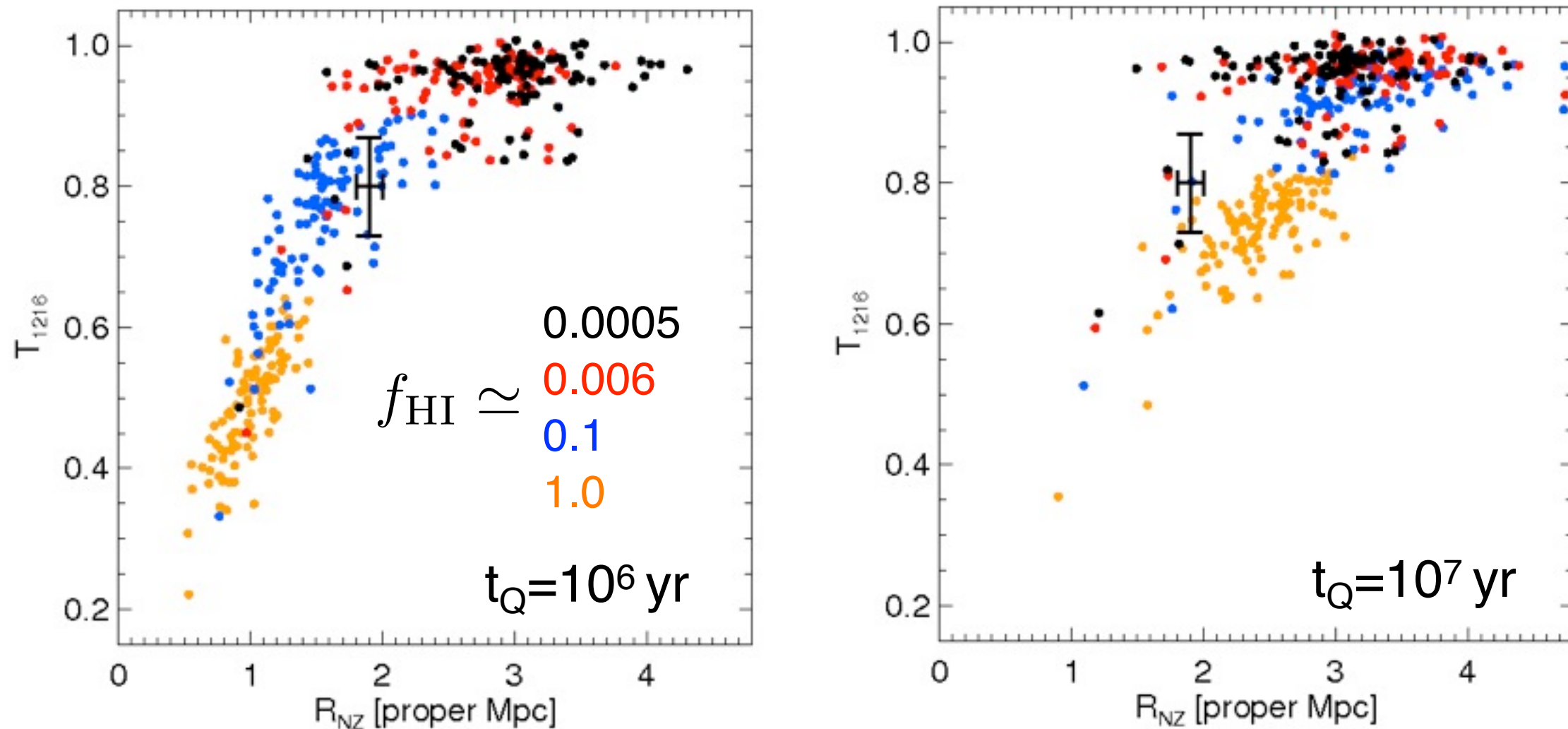
## 4) ULAS J1120+0641 at $z=7.085$



- **Small near-zone** (1.9 proper Mpc) for bright absolute magnitude ( $M_{1450}=-26.6$ )
- Spectrum also has smooth absorption component redward of the Ly- $\alpha$  emission line;
- Consistent with a **damping wing** from neutral IGM (e.g. Miralda-Escude 1998).

Mortlock et al. (2011)

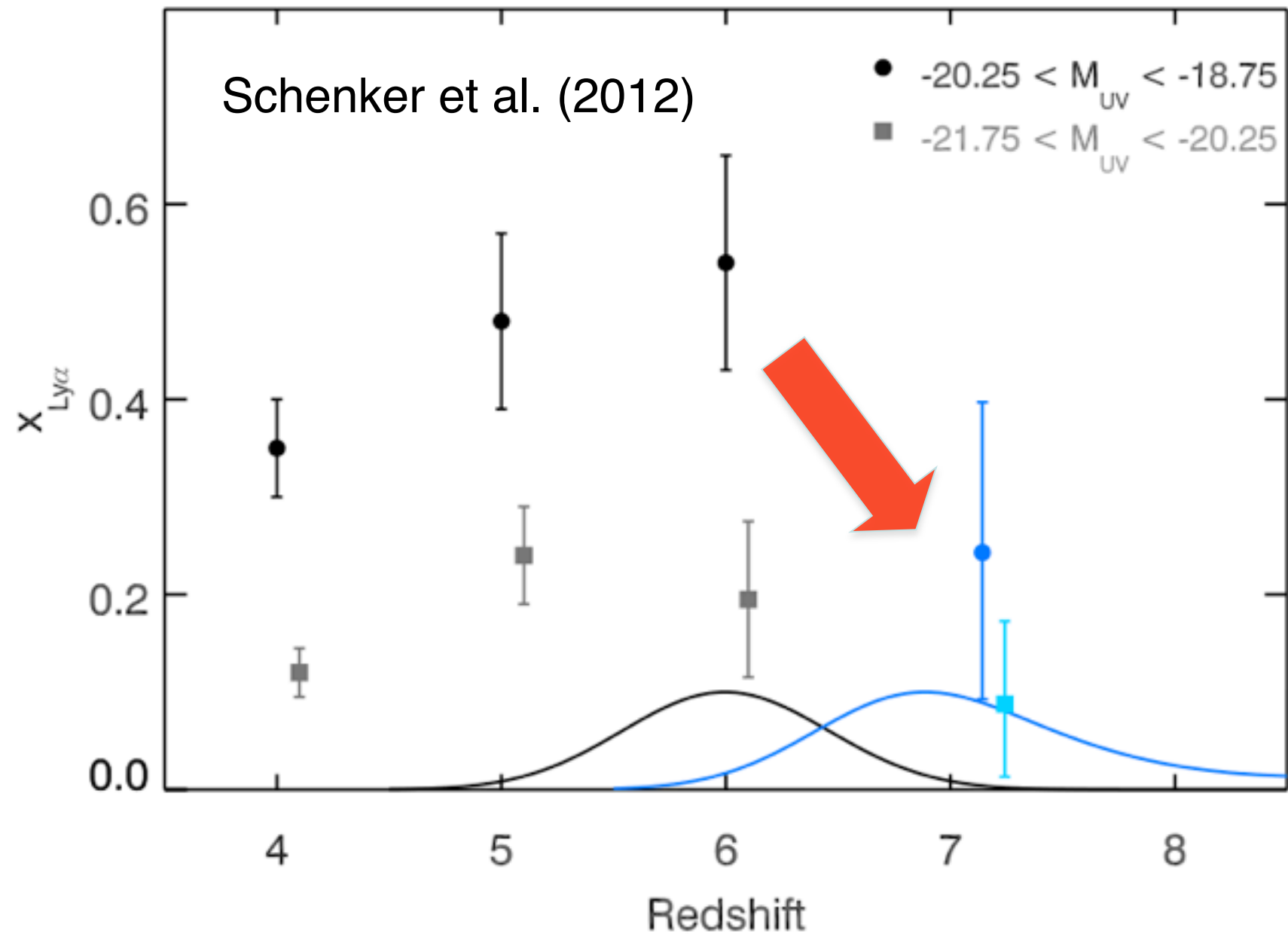
## 4) ULAS J1120+0641 at $z=7.085$



Bolton et al. (2011)

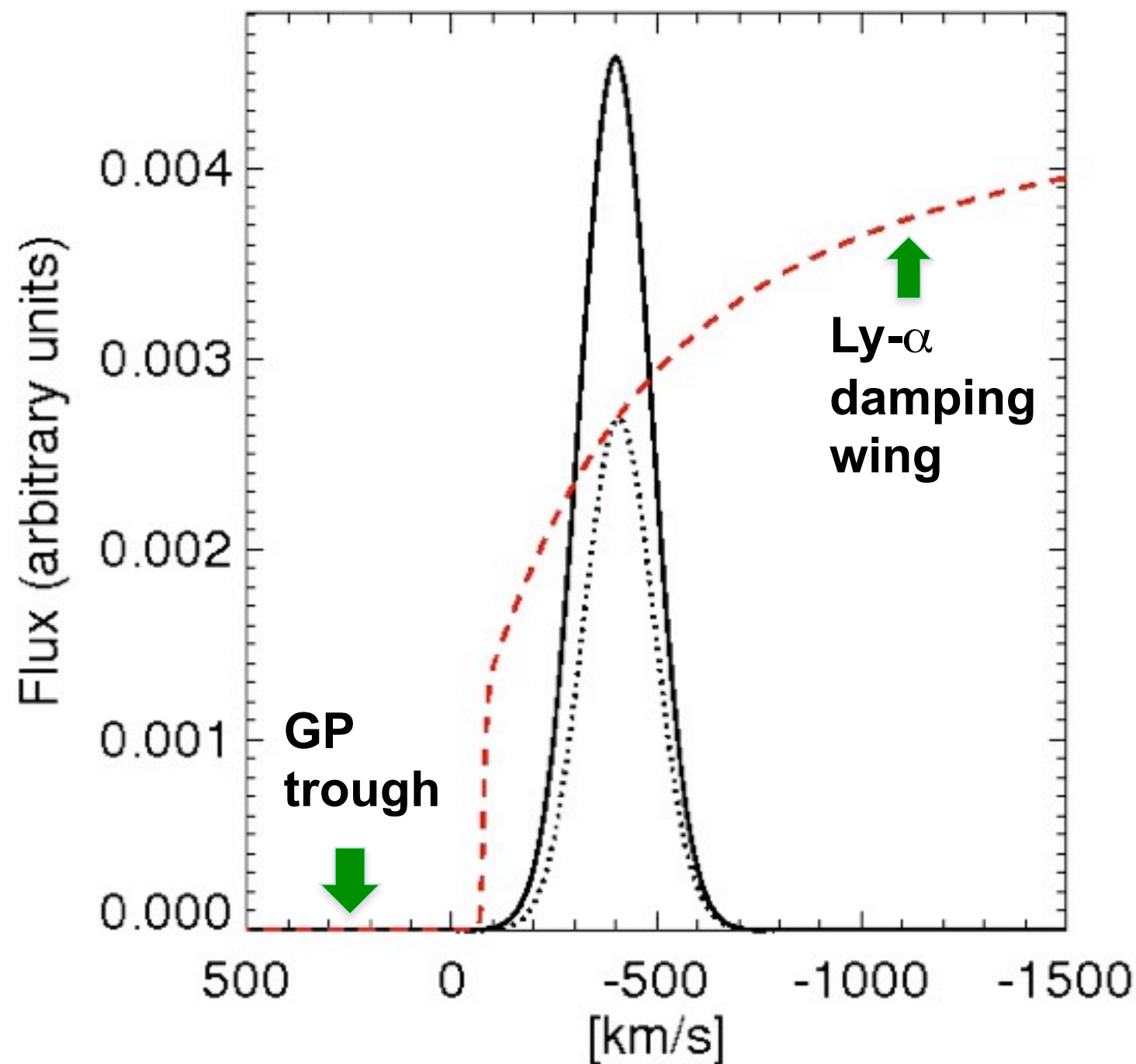
- For an optically bright phase of  $10^6 \text{ yr}$  ( $10^7 \text{ yr}$ ) the near-zone size/damping wing is consistent with  $f_{\text{HI}} \sim 0.1$  ( $\sim 1.0$ );

## 5) The LBG/LAE fraction



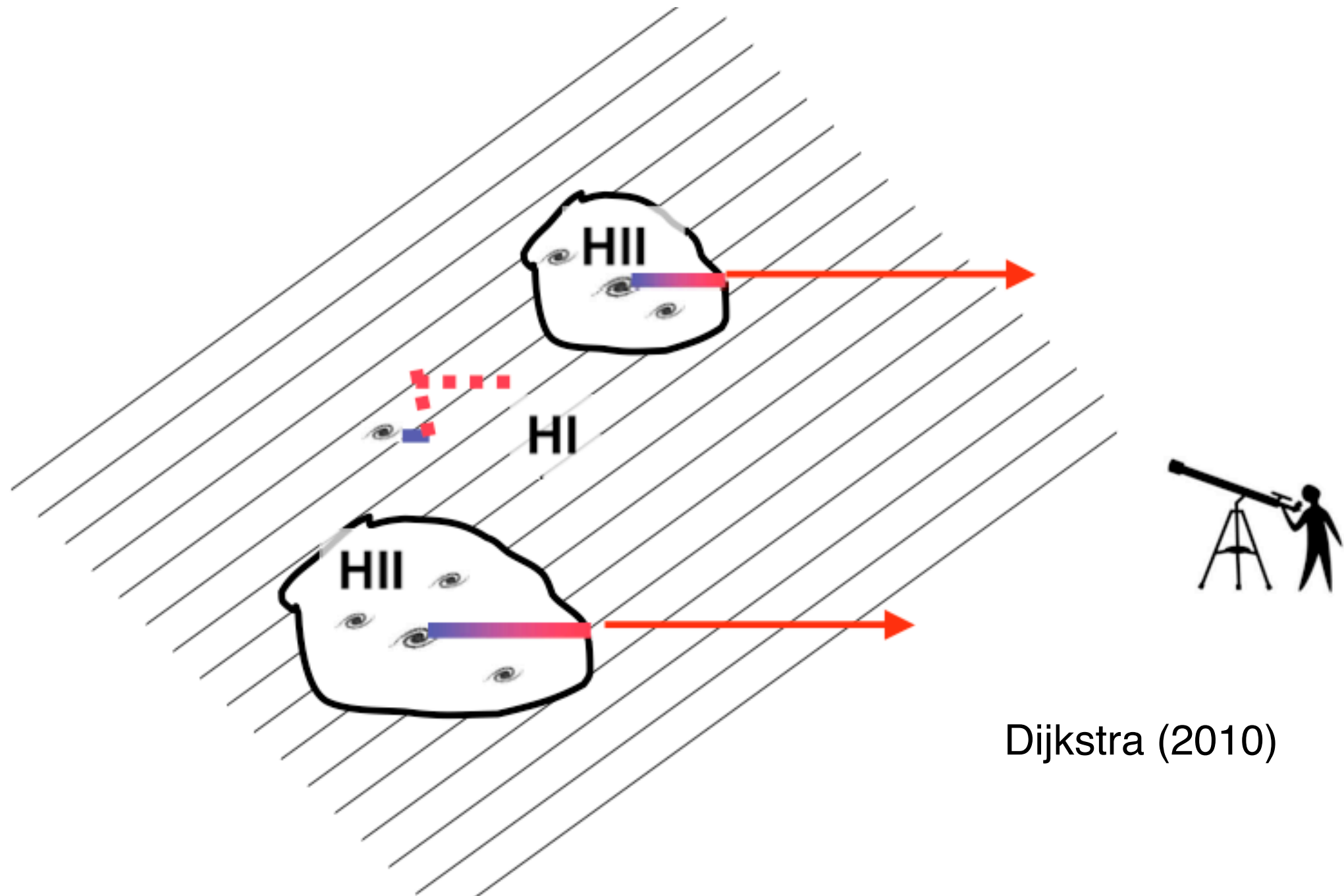
See also Pentericci+11, Vanzella+11, Ono+12, Curtis-Lake+12.

# The IGM damping wing



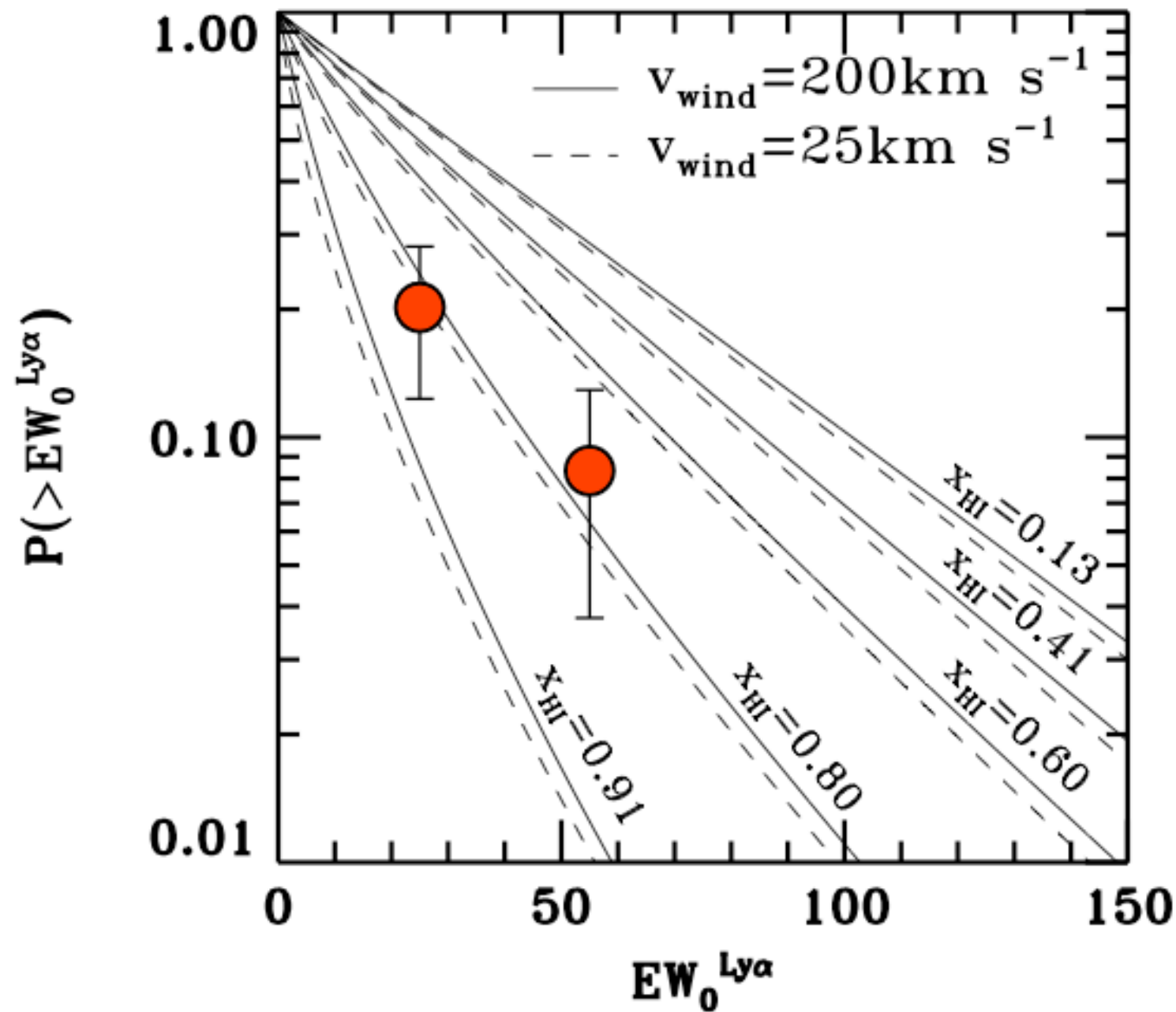
- Lorentzian wing suppresses Ly- $\alpha$  line if IGM significantly neutral, e.g. Miralda-Escude (1998);

# LAE visibility during reionisation



**But:** can still see Ly- $\alpha$  emission if the surrounding IGM is neutral - just need a big enough HII bubble + outflows which aid line scattering.

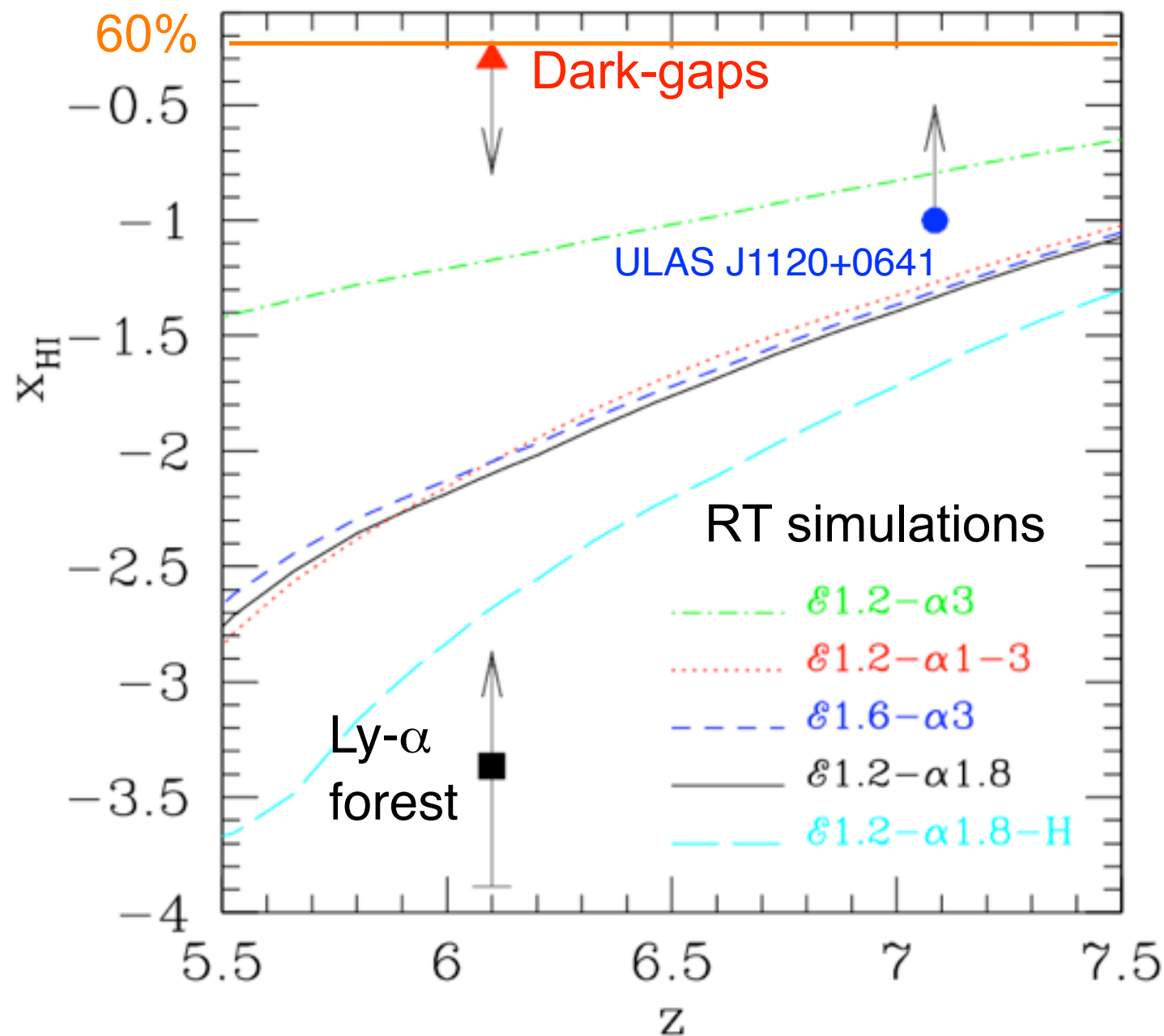
# Explaining the Ly- $\alpha$ EW distribution



- Very large HI fractions at  $z=7$  (40-90 per cent) may be required for EW evolution from  $z=6-7$ ;
- **Caution:** model assumes fully transparent IGM at  $z=6$  and no intrinsic LAE evolution.

Ono et al. (2012)  
see also Dijkstra et al. (2011)

# Tension with other constraints?



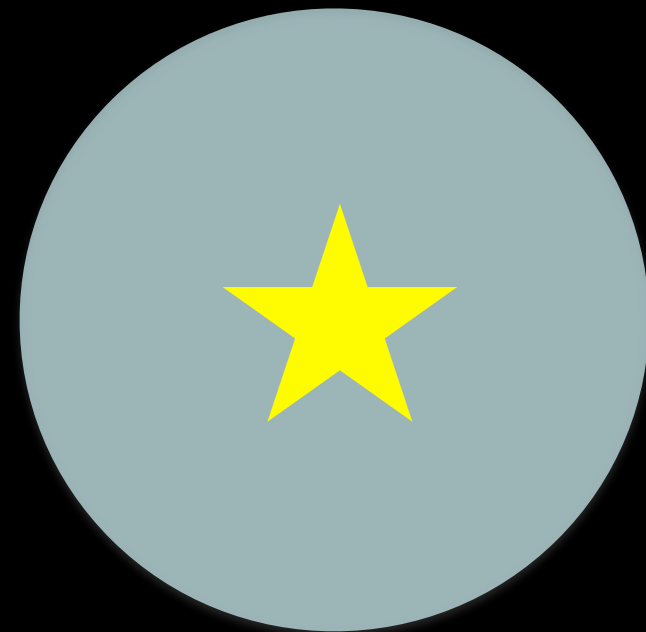
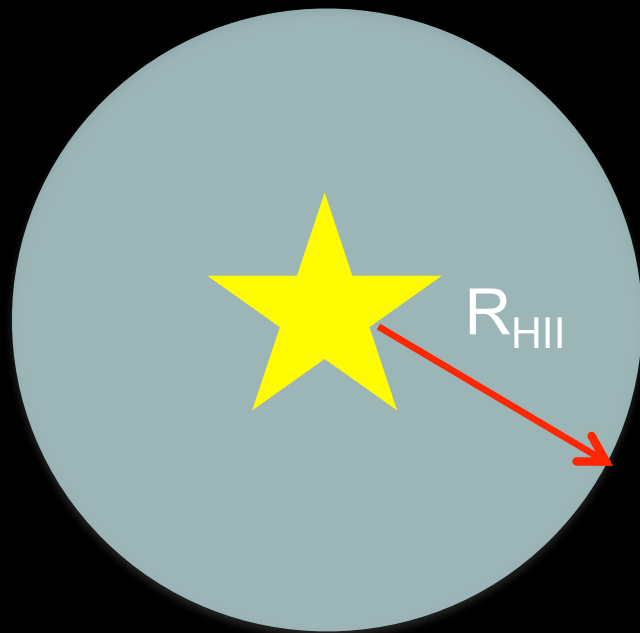
- Difficult to match other constraints (CMB, Ly- $\alpha$  forest emissivity) and simultaneously have  $f_{\text{HI}} \sim 0.6$  at  $z=7.1$ ;

Ciardi, JB et al. (2012)

Data from Fan et al. (2006), McGreer et al. (2011), Mortlock et al. (2012)

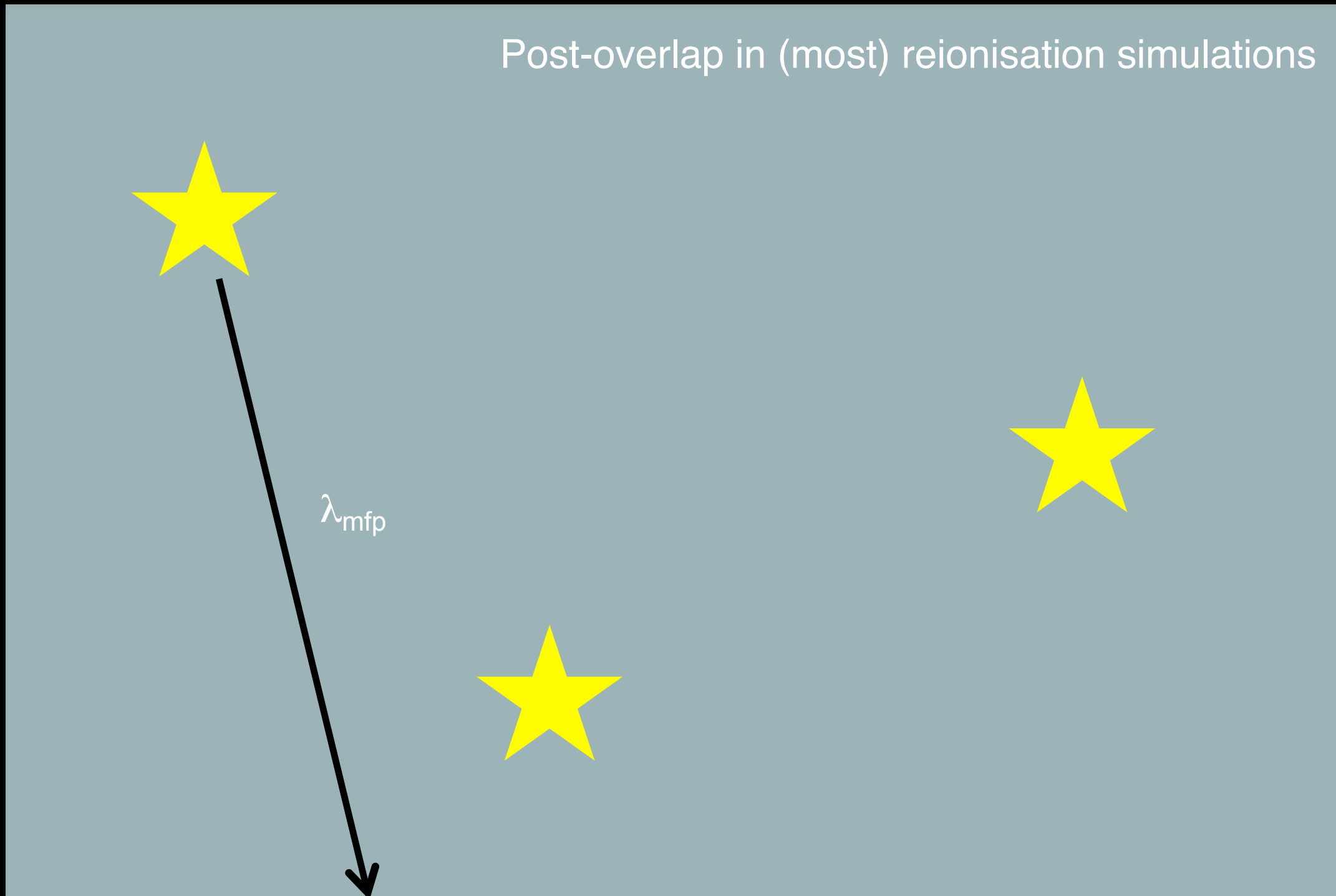
# Reionisation morphology

- “Pre-overlap”
- Mean free path for photons set by H-II bubble size

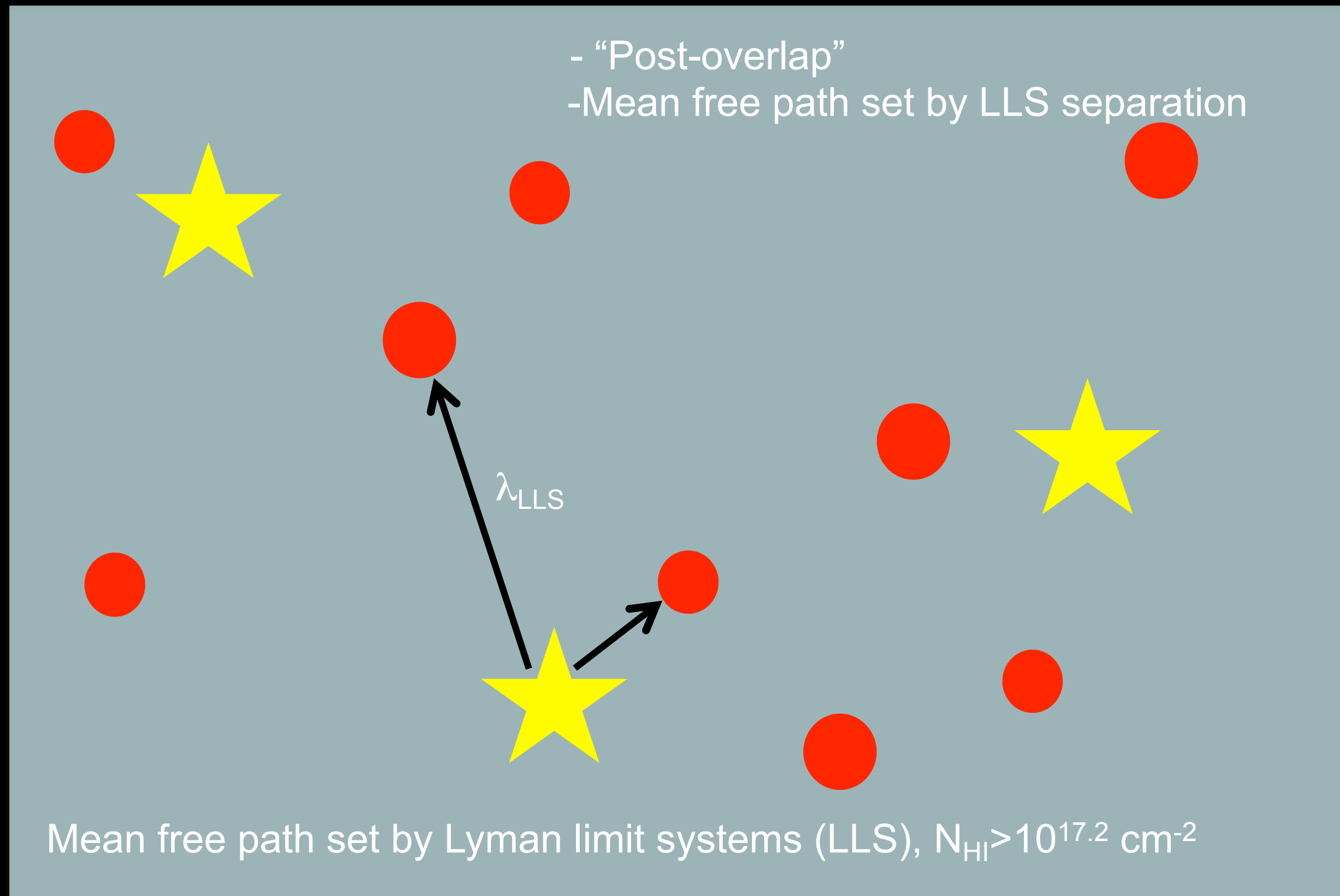


# Reionisation morphology

Post-overlap in (most) reionisation simulations



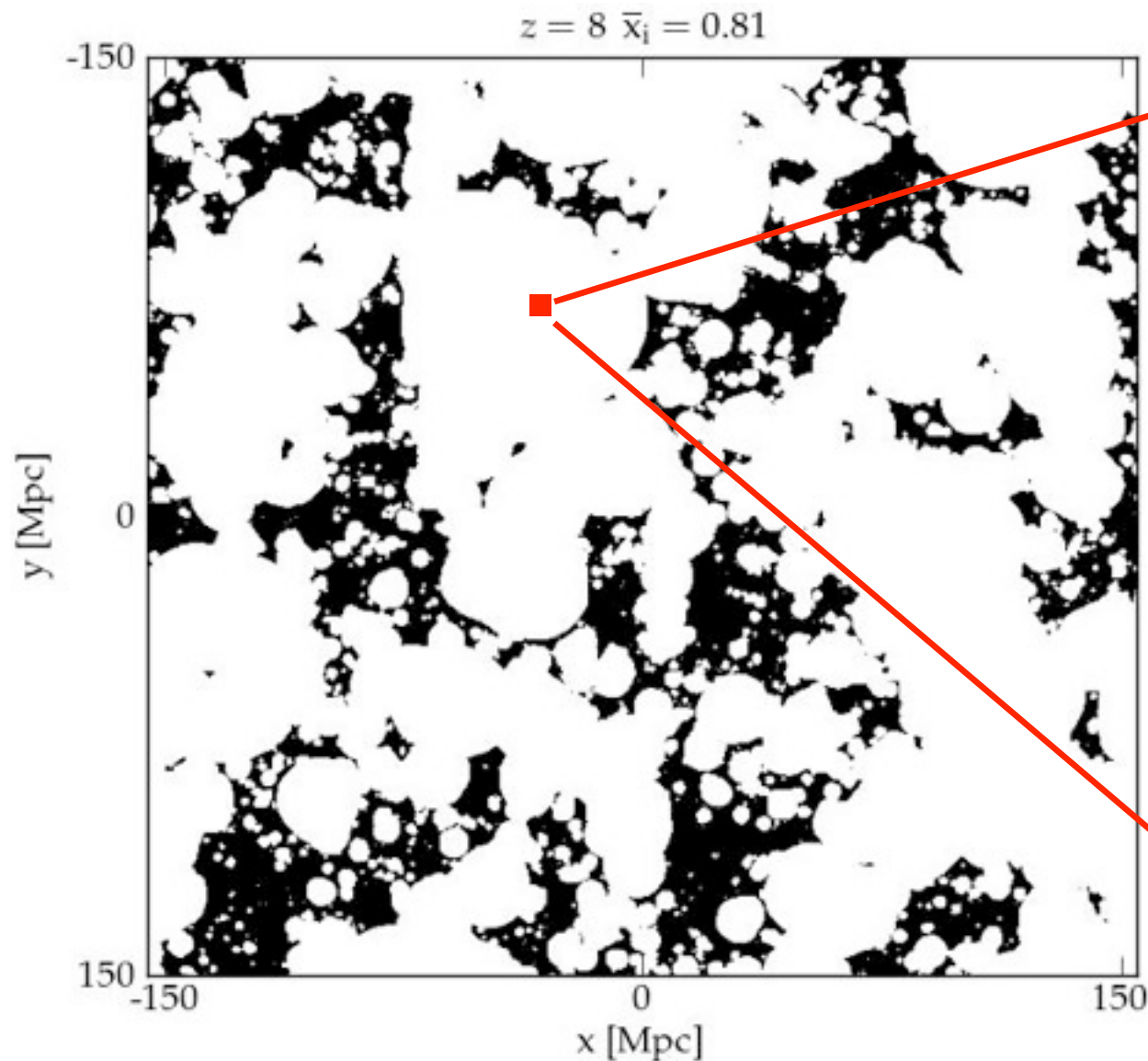
# Reionisation morphology



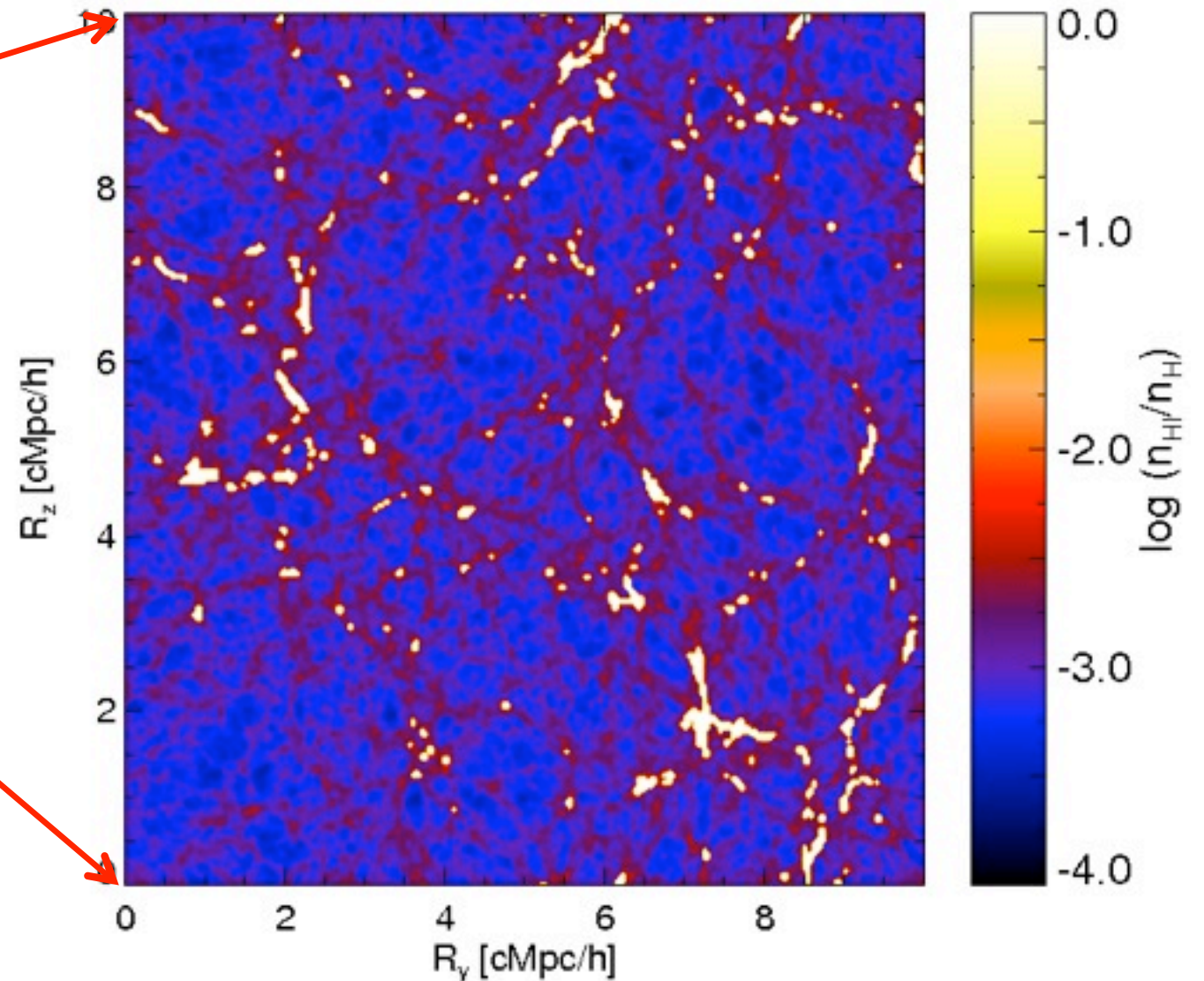
# An issue of dynamic range

Large scale simulations:  $L > 100$  cMpc.

Lyman-limit systems:  $L < 20$  pkpc

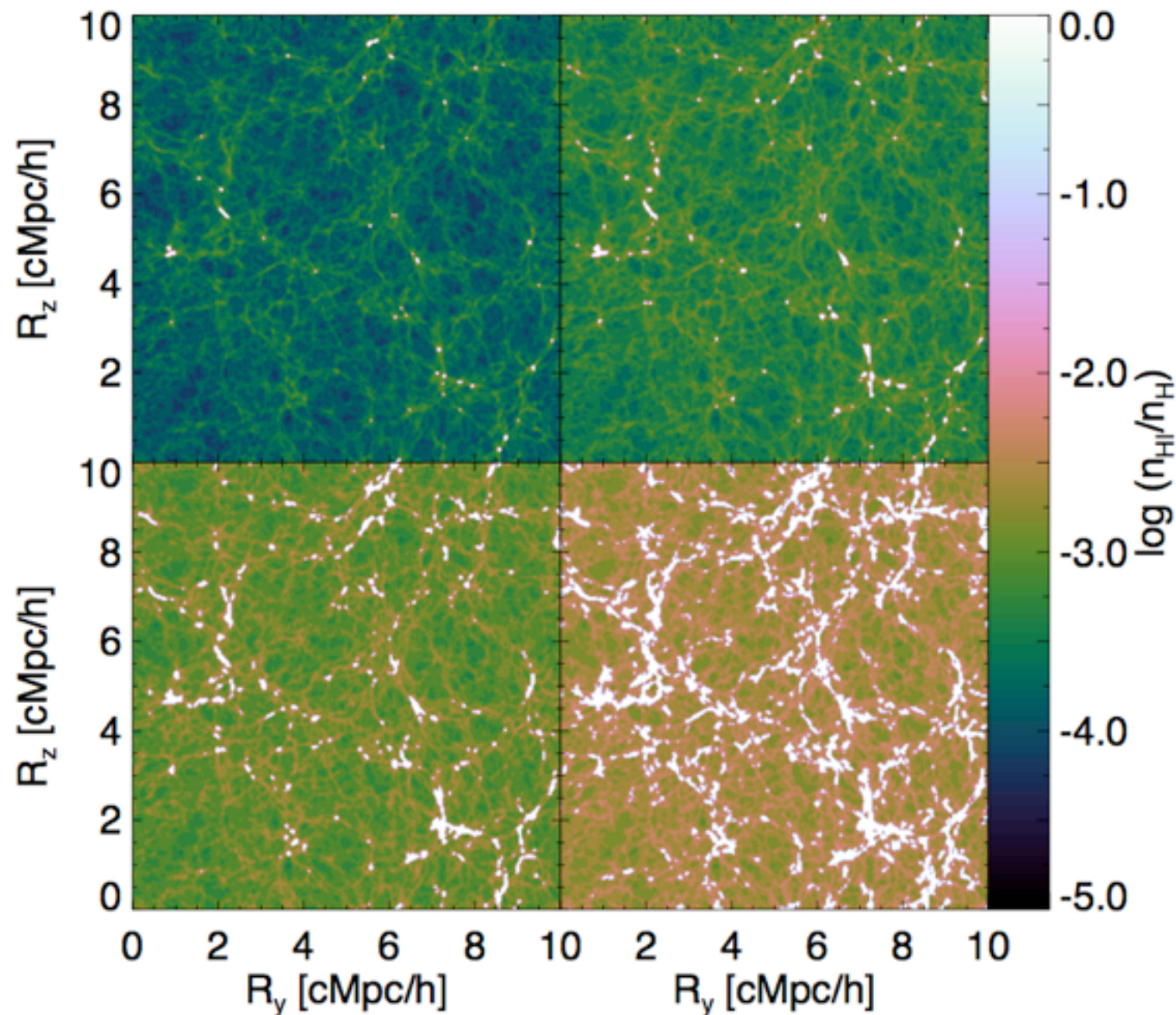


Large-scale reionisation simulation  
Santos et al. (2010)



Gadget-3 hydro simulation  
+ self shielding

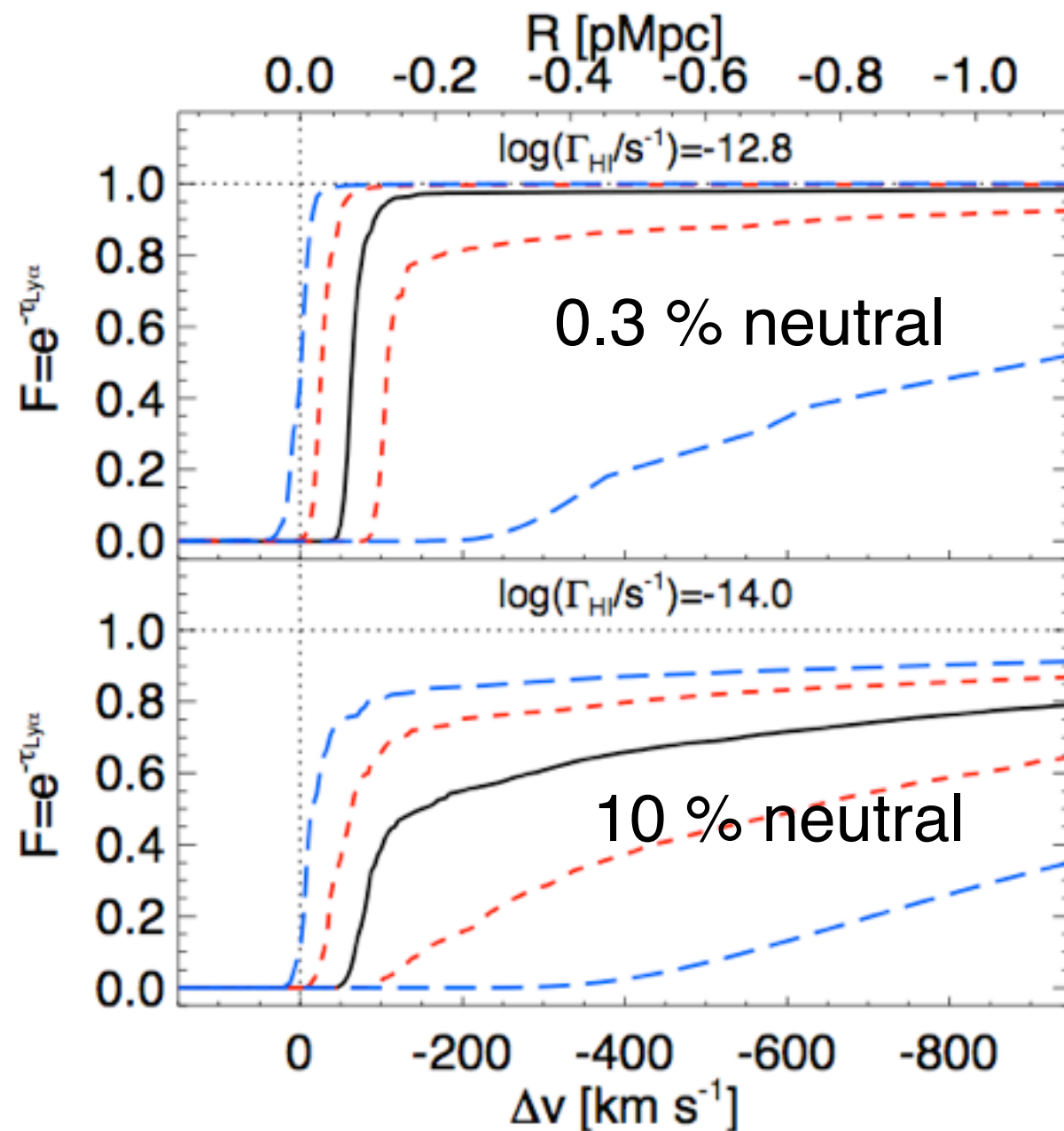
# Self-shielding approaching overlap



Bolton & Haehnelt (2013)

- Hydro simulation + self-shielding (see also McQuinn et al. 2011);
- Optically thick absorbers become larger and more numerous as the mean free path (and ionisation rate) decrease approaching overlap.

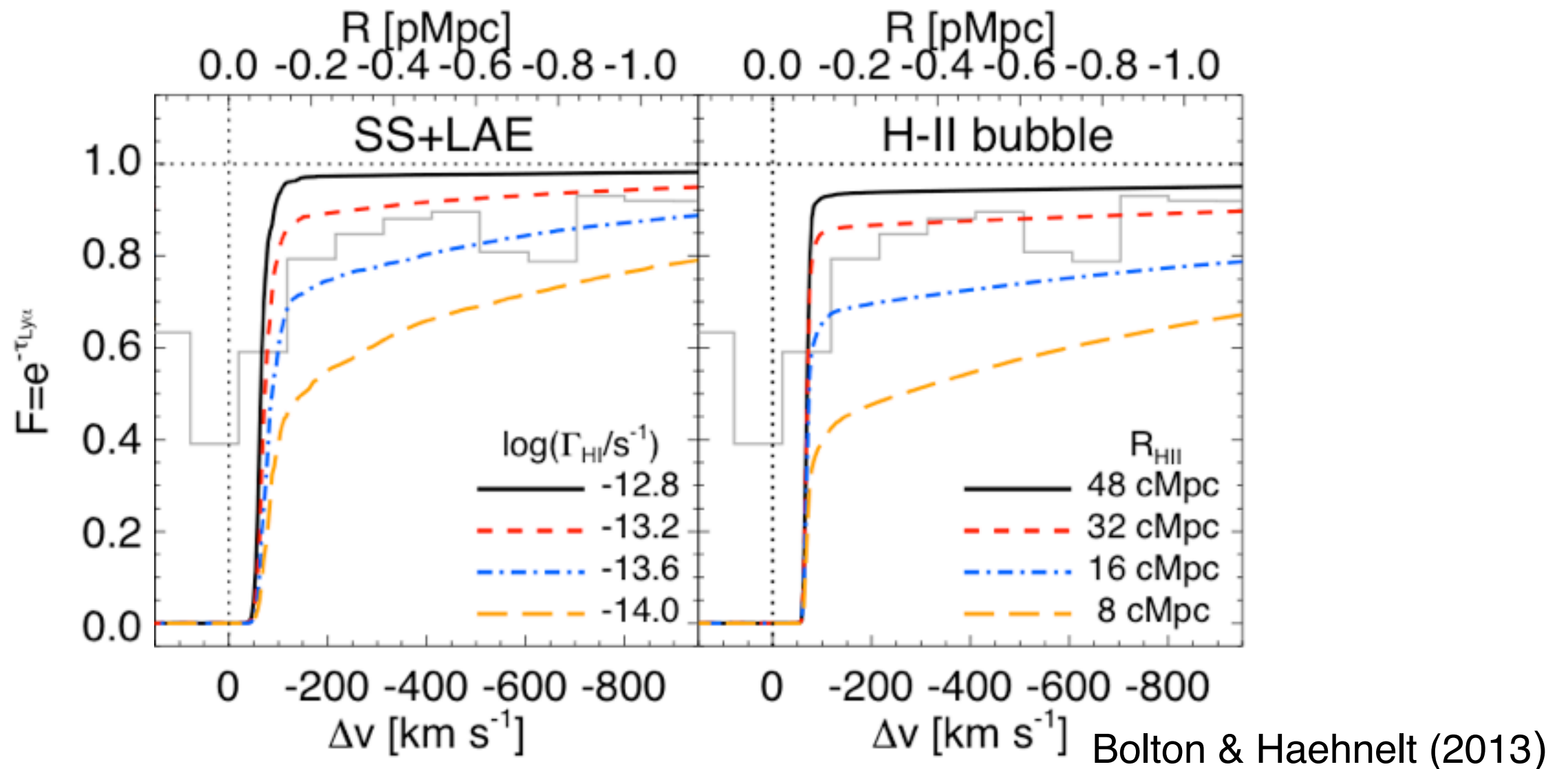
# Impact on the IGM damping wing



Bolton & Haehnelt (2013)

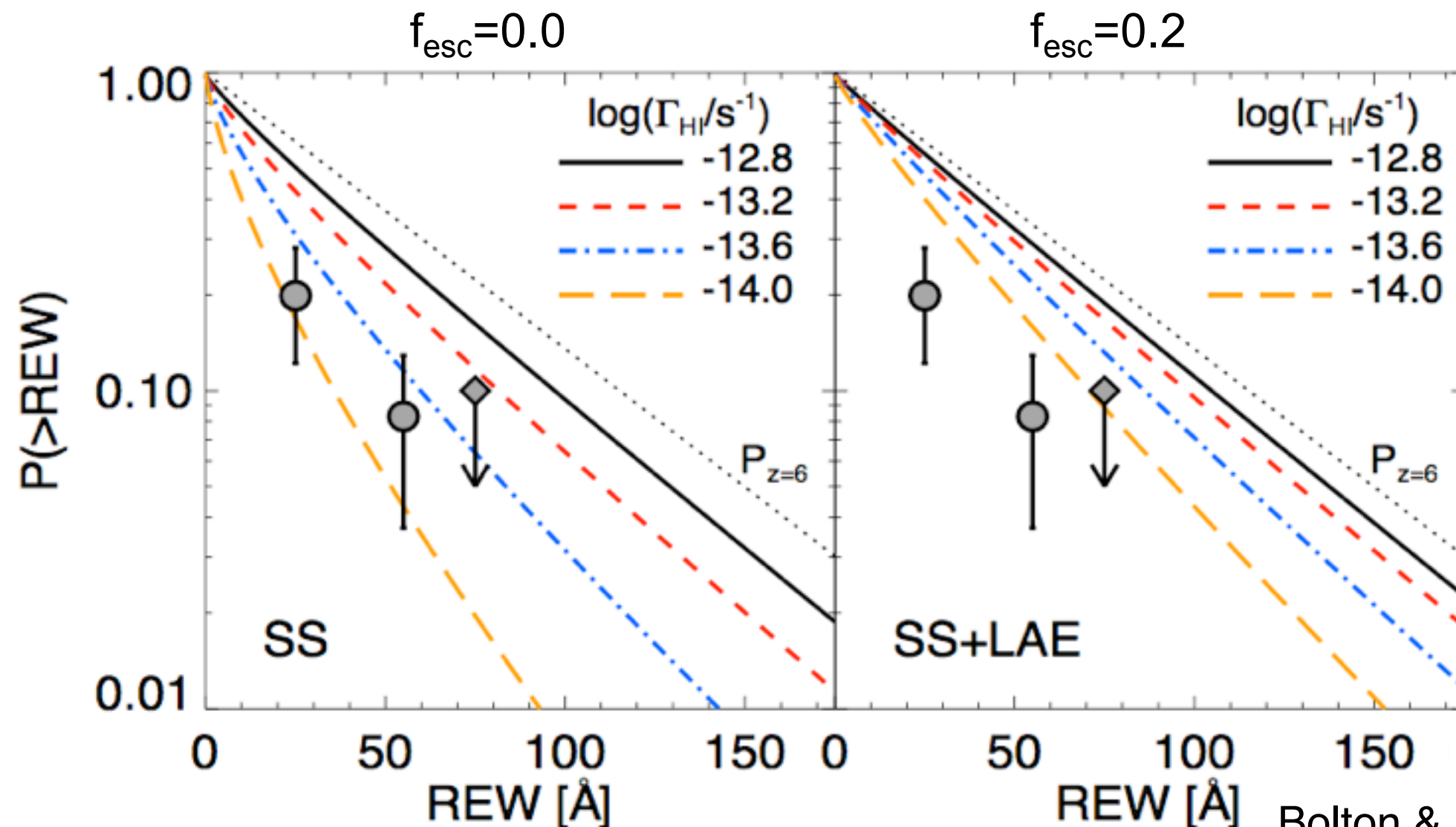
- Absorption from these systems varies significantly from one sight-line to another;
- High column density systems are vital for correctly estimating LAE visibility.

# Impact on the IGM damping wing



- Large scale reionisation simulations predict typical H-II region size at overlap is  $\sim 50$  cMpc/h, and  $\sim 30$  cMpc/h for 30 per cent neutral IGM, e.g. Zahn et al. (2011);
- Comparable red damping wing to an IGM which is 3% neutral when including optically thick absorbers.

# Equivalent width distribution

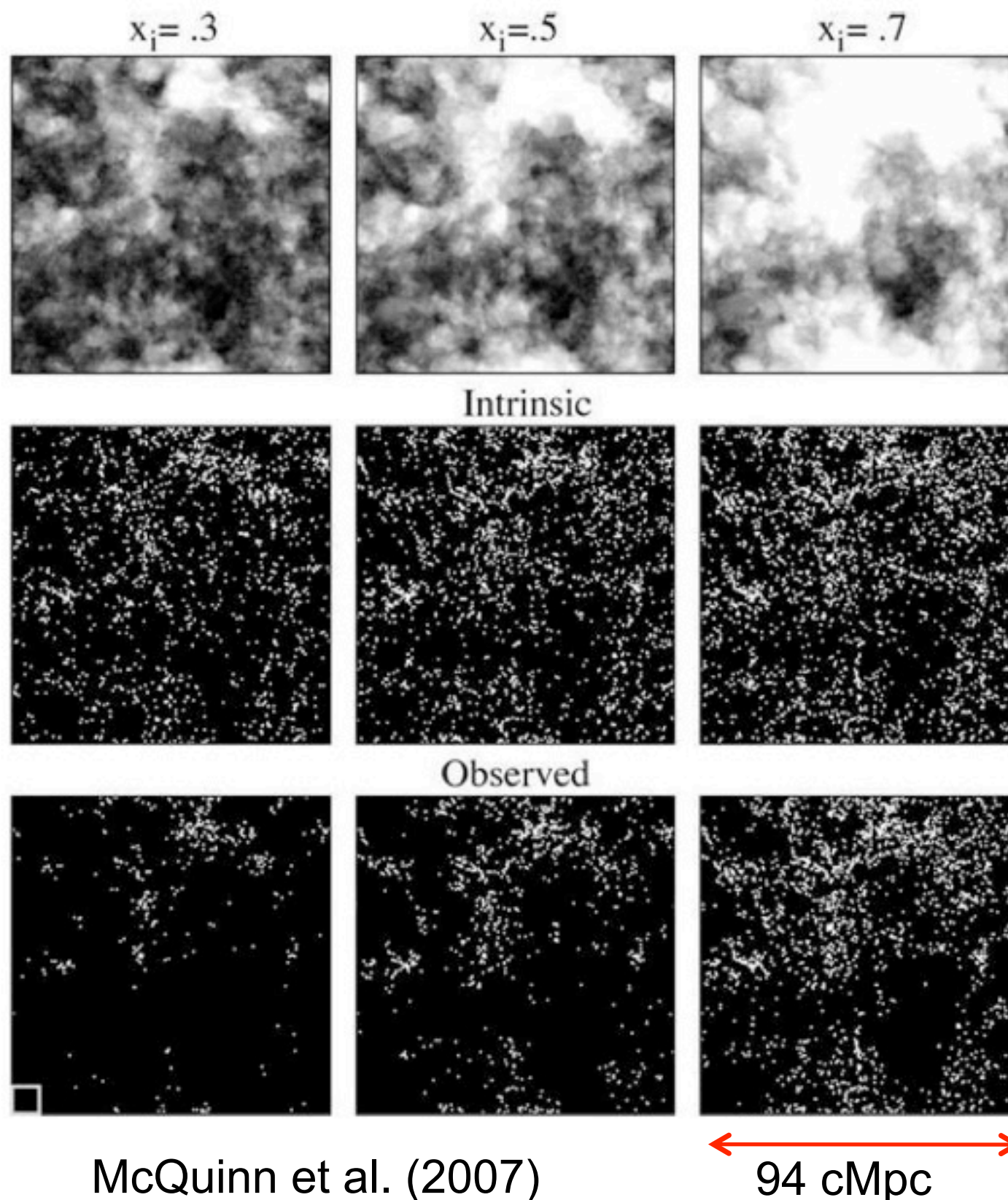


Bolton & Haehnelt (2013)

| $\log(\Gamma_{\text{HI}}/\text{s}^{-1})$ | $\langle f_{\text{HI}} \rangle_V$ |
|------------------------------------------|-----------------------------------|
| -12.8                                    | $2.7 \times 10^{-3}$              |
| -13.2                                    | $9.2 \times 10^{-3}$              |
| -13.6                                    | $3.2 \times 10^{-2}$              |
| -14.0                                    | $1.1 \times 10^{-1}$              |

Smaller neutral fractions account for data if LyC escape fraction from LAEs is low, partially alleviating tension with other constraints.

# Effect on Ly- $\alpha$ emitter clustering?



- LLS impact on the morphology of reionisation (e.g. Choudhury et al. 2009);
- Implications for the interpretation of forthcoming observations (e.g. High- $z$  Ly- $\alpha$  emitter clustering with HyperSuprimeCam)?

# Summary

- Current observational constraints on the timing and nature of reionisation are still weak, and not all are in agreement;
- Appears difficult to reconcile 60% neutral IGM at  $z=7$  from LBG/LAE fraction with other observational constraints (Ly- $\alpha$  forest, CMB, IGM temperature);
- Small-scale structure in the IGM (Lyman limit systems) may offer one route to (at least in part) resolving this. Resolving these systems is furthermore vital for modelling reionisation self-consistently;
- Further observations are key;  $z>7$  data are already helping to better inform theoretical models of reionisation. Imprint of LLS on clustering may be relevant for HSC.